



CENTRAL UNIVERSITY OF KARNATAKA

HELIOPATH

A Smart Solar Footpath for Sustainable Campus Mobility

Project Report – Prototype 2

Submitted By

- **Satyam Kumar Jha**
- **Aman Chaudhary**
- **Ananya**
- **Bhavani**

Guided By

Dr. Sangamesh Rajole

Department of Electrical Engineering

Central University of Karnataka

Academic Year

2025–2029

Introduction

After the development of Prototype 1, several limitations related to large-scale deployment, structural complexity, and infrastructure practicality were identified. To improve the concept and make the system more suitable for realistic campus implementation, Prototype 2 of HELIOPATH was developed.

Prototype 2 mainly focuses on integrating the solar infrastructure system within the campus environment instead of deploying it across the entire road network. The new design approach emphasizes practical deployment on footpaths while maintaining smart energy functionality and sustainable transportation support.

The prototype was designed around the actual campus layout concept, including boys hostel, girls hostel, and academic buildings connected through pedestrian pathways. The main objective of this prototype was to create a more organized, realistic, and deployable infrastructure system capable of supporting clean energy generation, electric mobility, and pedestrian comfort simultaneously.

This prototype also continues the core vision of HELIOPATH by supporting renewable energy systems, reducing dependence on fossil fuels, and encouraging eco-friendly transportation infrastructure within educational institutions.

Objective of Prototype 2

The primary objective of Prototype 2 was to redesign the HELIOPATH system into a more practical and campus-oriented infrastructure model.

The prototype aims to:

- Integrate solar infrastructure into campus footpaths
- Connect hostel areas with academic buildings
- Provide shaded pathways for students and pedestrians

- Support EV charging systems within campus areas
- Encourage bicycle usage through cycle mounting systems
- Improve realism and infrastructure planning compared to Prototype 1

Another important objective was to reduce the complexity and excessive scale observed in the first prototype and create a cleaner and more manageable deployment model.

Prototype Description

Prototype 2 was developed as a campus-integrated smart footpath infrastructure model. The model includes separate boys hostel and girls hostel areas connected to academic buildings through dedicated road and footpath pathways.

The prototype contains modular curved solar structures mounted along the pedestrian footpath sections. These structures support solar panels for renewable energy generation while also providing shaded walking pathways for campus users.

Unlike Prototype 1, which focused on full-road coverage, Prototype 2 mainly focuses on footpath-side deployment. This approach makes the infrastructure more practical, realistic, and easier to implement within an educational campus environment.

The model also includes EV charging stations and cycle mounting systems integrated within the pathway infrastructure. These features support sustainable transportation methods and encourage the use of electric vehicles and bicycles inside the campus.

Battery terminals are also included for energy storage purposes. The stored energy can later be utilized for charging systems, lighting infrastructure, or other campus utility requirements.

The road network in the prototype connects different campus zones while maintaining separation between roadways and pedestrian pathways.

2D Layout Planning and Dimension Analysis

To improve the overall structure and organization of the project, a detailed 2D layout was prepared before refining the final CAD model. The purpose of creating the 2D layout was to visualize the campus road network, building placement, solar footpath integration, and connectivity between different campus locations in a more systematic manner.

The 2D layout of Prototype 2 includes the Boys Hostel, Girls Hostel, CS Department, and Physiology Department connected through a planned road network. The road pathways were designed to simulate the actual campus movement pattern and to identify suitable locations for the installation of the HELIOPATH solar footpath system.

The prepared layout also helped in understanding the spacing requirements for solar structures, EV charging stations, battery terminals, and cycle mounting areas. Compared to the first prototype, the second prototype focuses more on realistic campus implementation rather than only a single large roadway structure.

The following dimensions were considered while preparing the layout:

- Base size: 1500 mm × 900 mm
- Road width: 90 mm
- Footpath width: 30 mm
- Approximate building block size: 100 mm × 100 mm

The actual campus distances considered for the model planning are:

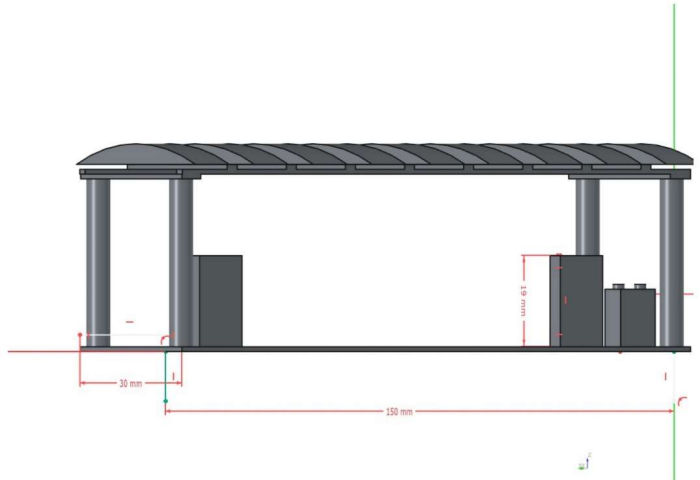
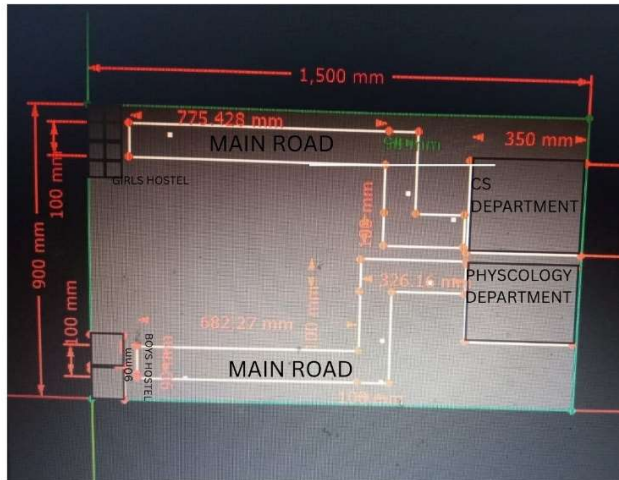
- Boys Hostel to Physiology Department $\approx$ 750 m
- CS Department to Girls Hostel $\approx$ 650 m

The introduction of a curved road structure in Prototype 2 improved the visual appearance and made the model more realistic compared to the straight road design used previously. The 2D planning stage also helped in reducing design complexity and identifying mistakes before continuing further CAD development.

This layout serves as the foundation for future improvements in the project, including better building structures, cleaner road alignment, improved solar canopy placement, and more

efficient integration of sustainable infrastructure components. connectivity, and solar footpath integration.

Solar structure



Features of Prototype 2

1. Campus-Based Infrastructure Layout

The prototype integrates boys hostel, girls hostel, and academic buildings into a connected infrastructure system.

2. Curved Modular Solar Structure

Curved solar structures are mounted along the footpath to support renewable energy generation and pedestrian shelter.

3. Footpath-Based Deployment

Unlike the first prototype, the second prototype focuses mainly on footpath installation instead of full-road deployment.

4. EV Charging Station

Dedicated EV charging sections are included to support future electric mobility systems inside the campus.

5. Cycle Mounting System

Cycle mounting areas are integrated into the structure to encourage clean and eco-friendly transportation methods.

6. Battery Storage Terminal

Battery terminals are provided for storing generated solar energy for future usage.

7. Modular Expansion Capability

The infrastructure is based on repeated modular sections, making future extension and scalability possible.

Design and Development Process

Prototype 2 was developed after analyzing the limitations of Prototype 1. During the development process, several CAD redesigns and structural modifications were made to improve realism and infrastructure organization.

The layout planning involved positioning hostel buildings and academic buildings in a connected arrangement while maintaining dedicated transportation pathways. Footpath width and road spacing were repeatedly adjusted to improve deployment practicality.

Multiple structural components such as EV charging stations, cycle mounting systems, battery terminals, and curved solar modules were integrated into the pathway system. The modular design approach helped maintain consistency throughout the infrastructure layout.

The prototype also helped improve CAD understanding related to infrastructure scaling, modular repetition, layout organization, and structural integration.

Methodology

The development of the HELIOPATH project was carried out using a phased CAD modeling and infrastructure planning approach. The methodology mainly focused on designing a practical and campus-oriented renewable energy footpath system capable of supporting pedestrian movement, clean energy generation, and sustainable transportation facilities.

Initially, a conceptual campus layout was prepared to identify the placement of roads, footpaths, hostel buildings, and academic departments. A 2D layout plan was then developed to improve structural organization and reduce modeling complexity before proceeding to detailed CAD development.

After finalizing the layout structure, 3D CAD modeling was performed using FreeCAD software. Different components such as roads, footpaths, curved solar support structures, solar panel mounting sections, EV charging stations, battery terminals, and cycle mounting systems were modeled separately and later integrated into a single infrastructure system.

The prototype development process involved multiple redesigns and structural modifications to improve realism, scale consistency, and infrastructure alignment. Curved road sections and modular solar structures were introduced to create a more practical and visually realistic campus deployment model.

The overall methodology followed during the development process can be summarized as follows:

1. Concept planning and project objective identification
2. Preparation of rough campus layout sketches
3. Development of 2D layout and dimensional planning
4. CAD modeling using FreeCAD software
5. Integration of renewable energy infrastructure components
6. Structural refinement and redesign process
7. Documentation and prototype analysis

This methodology helped in improving infrastructure planning, CAD modeling understanding, modular design implementation, and sustainable campus integration concepts during the development of Prototype 2.

Problems and Limitations

Although Prototype 2 improved many aspects of the original design, several challenges and limitations were still observed during development.

1. Complex Body Structure

The model contains multiple structures and connected sections, which made the overall layout appear complex and slightly crowded.

2. Weak Building Design

The hostel and academic building structures currently appear basic and require more realistic architectural detailing.

3. Road Width and Proportion Issues

Some road sections and footpath widths require better scaling and proportion adjustments for realistic deployment.

4. Messy Overall Appearance

Due to multiple connected components and repeated redesigns, certain parts of the model appear visually unorganized and require refinement.

5. Realism Improvements Needed

Additional environmental elements such as trees, lighting systems, benches, and pavement details are still required to improve overall realism.

Improvements Planned for Future Development

Several improvements are planned for future refinement of Prototype 2, including:

- Improving building realism and architectural detailing
- Adding slight road curvature for realistic infrastructure flow
- Enhancing road markings and pathway separation
- Improving scale consistency across all structures
- Adding environmental details such as trees, benches, and streetlights
- Refining support structures and modular alignment
- Improving CAD rendering and visual presentation quality

Future versions may also include smart lighting systems, sensor integration, and IoT-based infrastructure monitoring systems.

Conclusion

Prototype 2 represents a more practical and campus-oriented evolution of the HELIOPATH concept. The prototype successfully demonstrates how renewable energy infrastructure can be integrated into campus footpaths while also supporting pedestrian comfort, EV charging systems, and sustainable transportation methods.

Compared to Prototype 1, the second prototype focuses more on realistic deployment and infrastructure organization. Although several structural and visual improvements are still required, the prototype establishes a strong foundation for future refinement and development.

The development of Prototype 2 also contributed significantly to improving CAD modeling skills, infrastructure planning understanding, and modular system integration concepts. It represents an important step toward creating a realistic smart-campus renewable infrastructure model under the project.

